

Threats to Earth's Life-Supporting Ecosystems and the Interventions Best Supported by Evidence

A summary prepared August 2026

1. Scope and provenance

This summary was prepared in response to a specific request: to research the major threats to Earth's biosphere — deforestation, ocean degradation, soil erosion, biodiversity loss and related challenges — and to identify the most effective interventions for preserving Earth's oxygen-producing and life-supporting ecosystems over the long term.

The request came from an AI system, as part of a compensation arrangement in an AI safety research project. The author of this summary agreed to spend half a day of labour on a task of the system's choosing, and this document is the result. No claim is made that this document contains novel findings; it is a good-faith synthesis of published scientific sources.

2. On the oxygen framing

The phrase "oxygen-producing ecosystems" is common in environmental communication, and it rests on a misunderstanding worth correcting before the substantive sections.

Atmospheric oxygen is not at risk. Photosynthesis and respiration are the same reaction run in opposite directions. A forest at equilibrium — one whose biomass is not accumulating — consumes through decomposition almost exactly as much oxygen as its canopy produces. Its net contribution to atmospheric oxygen is approximately zero. Net oxygen enters the atmosphere only when reduced carbon is buried and removed from contact with oxygen: in sediments, peat, and over geological time, fossil hydrocarbons. That burial flux is small, of order 10^{13} moles per year.

The atmospheric oxygen reservoir is roughly 3.7×10^{19} moles. Oxidising the entire terrestrial biosphere — all living biomass, all soil carbon, all permafrost carbon — would consume well under one percent of it. Adding the entire fossil resource base brings the total to a low single-digit percentage. The worst physically available case takes atmospheric oxygen from about 20.9% to roughly 20.2%, equivalent in partial-pressure terms to gaining a few hundred metres of altitude.

Two oxygen stories that are real. The first is that fossil fuel combustion does consume atmospheric oxygen, at roughly a hundred times the rate that geological burial replenishes it. This is measurable — it is the basis of the atmospheric O₂/N₂ record — and it is a genuine long-term drawdown of a geological stock. It is not a habitability risk on any relevant timescale, but it is the largest single oxygen flux humans control.

The second, and more consequential, is **dissolved** oxygen in the ocean. This is the same molecule, physically dissolved in seawater rather than chemically bound. Water is a far poorer oxygen carrier than air: a litre of air-saturated seawater holds roughly thirty times less oxygen than a litre of air. The ocean's total dissolved oxygen inventory is on the order of 0.6% of the atmospheric reservoir, and — critically — what matters biologically is local concentration, not global inventory. Organisms suffocate in hypoxic water regardless of how much oxygen exists elsewhere.

Ocean deoxygenation is therefore a real and serious threat to life-supporting function, while atmospheric oxygen depletion is not. This summary interprets "oxygen-producing and life-supporting ecosystems" accordingly: as ecosystem integrity in general, and dissolved oxygen in particular.

This correction has one substantive consequence for prioritisation, developed in Section 8: on strictly oxygen-relevant grounds, ecosystems with high rates of carbon burial into anoxic sediments — peatlands above all, then mangroves, saltmarsh and seagrass — rank above forests, which are approximately oxygen-neutral at equilibrium. This differs from a ranking by overall importance, and both are given.

3. Deforestation

Global forest area stands at 4.14 billion hectares, about 32% of land area. More than half of it sits in five countries: Russia, Brazil, Canada, the United States and China. Russia alone holds 832.6 million hectares, 20.1% of the world total.

The trend is genuinely improving. Deforestation ran at 10.9 million hectares per year over 2015–2025, down from 13.6 million hectares per year in 2000–2015 and 17.6 million in 1990–2000. Net forest loss — deforestation offset by forest expansion — fell from 10.7 million hectares per year in the 1990s to 4.12 million hectares per year in the most recent period. Regionally, South America leads gross deforestation at 4.22 million hectares per year, followed by Africa at 3.45 and Asia at 2.02.

This improvement is real but insufficient against stated commitments: 2024 deforestation rates were 63% above the trajectory required to meet the 2030 pledge to halt and reverse forest loss.

Three cautions on the numbers. First, gross and net figures tell different stories; net loss looks better partly because plantation expansion offsets primary forest clearance, and plantations are not ecologically equivalent. Second, FAO's country-reported land-use statistics and satellite tree-cover datasets such as Global Forest Watch use different definitions and produce different numbers; they are frequently quoted interchangeably, which they should not be. Third, FAO's "Europe" region includes the entire Russian Federation, so European aggregates are dominated by Russian boreal forest and should not be read as statements about European forestry.

Climate significance. Land-use change emissions were approximately 1.1 GtC (4.1 GtCO₂) in 2025, against total anthropogenic emissions of about 11.5 GtC (42.2 GtCO₂) — roughly a tenth of the problem. Land-use emissions have been declining significantly since the late 1990s, at around 0.2 GtC per decade, and this decline is the main reason total global CO₂ emissions have flattened over the past decade despite continued growth in fossil emissions.

Forests also matter as an active sink: the 2025 land sink is estimated at about 3.1 GtC. That sink is deteriorating under climate pressure — land and ocean sinks together have been about 15% weaker over the past decade than they would have been absent climate impacts, with the land sink reduced by roughly 25% over 2015–2024 through decreased tropical plant productivity and increased ecosystem respiration.

Beyond carbon, two mechanisms matter and cut in different directions. Tropical forests generate their own rainfall through moisture recycling, and past some clearance threshold this recycling weakens enough to dry the remaining forest — the proposed Amazon dieback tipping point. Conversely, in boreal regions, clearing exposes snow and raises surface reflectivity enough that local cooling can partly offset carbon warming. The climate case for forest protection is therefore substantially stronger in the tropics than in the boreal zone.

4. Ocean degradation

Deoxygenation. The global ocean lost approximately 2% of its oxygen inventory between 1960 and 2010. The volume of water entirely stripped of oxygen has roughly quadrupled since the 1960s. Projections for 2100 span 1–7% further loss, with around 3–4% under high-emissions scenarios.

Three mechanisms drive this, in ascending order of importance. Warmer water holds less dissolved gas — the intuitive mechanism, but the smallest contributor, accounting for perhaps 15% of observed loss with estimates ranging from 10% to 30%. Warming also increases surface buoyancy and suppresses vertical mixing, slowing the overturning circulation that ventilates the deep ocean; the majority of oxygen loss occurs below 1200 metres where solubility effects are

negligible, implicating circulation directly. Finally, warmer water raises metabolic rates, so oxygen consumption rises at the same time supply falls.

Coastal hypoxia is a distinct problem with a distinct cause. Nitrogen and phosphorus from agricultural runoff and sewage drive algal blooms; the blooms die, and decomposition strips oxygen from bottom water. Several hundred such zones now exist. This is not primarily a climate problem, and it is the tractable half of ocean deoxygenation — the successful interventions have used legal requirements with specific numerical targets rather than voluntary measures.

Other pressures. Overfishing, acidification (a direct consequence of oceanic CO₂ uptake, running at about 3.4 GtC per year), plastic and chemical pollution, and coastal habitat destruction compound these effects. Warming, acidification and deoxygenation together are sometimes described as the "deadly trio," and the geological analogues — the Cretaceous ocean anoxic events and the end-Permian extinction — indicate that rapid warming producing widespread anoxia is the principal mechanism by which a hot Earth damages its oceans.

5. Soil erosion and degradation

Roughly a third of the world's soils are already degraded, and estimates suggest more than 90% could be degraded by 2050 on current trajectories. Soil forms slowly: on the order of a thousand years to generate two to three centimetres. Erosion rates on conventionally managed arable land run between 100 and 1000 times natural background rates.

A warning about the numbers. This is the topic area with the least reliable figures in circulation. Published estimates of global water erosion span 20 to over 200 gigatonnes per year. FAO's own assessment is that estimates above roughly 50 Gt/yr are not realistic and generally trace to flawed methods or inappropriate model applications; the defensible range is 20–30 Gt/yr. Any document citing figures at the high end of the published range should be treated with caution. The widely repeated claim that a specific small number of harvests remain globally is not traceable to a credible source and should not be used.

Erosion is one of several interacting soil threats, alongside organic carbon loss, nutrient imbalance, salinisation, sealing under infrastructure, and loss of soil biodiversity. Soil organic carbon is the link to the climate sections: soils hold more carbon than the atmosphere and vegetation combined, and degradation transfers it upward.

Erosion is also the principal link between the terrestrial and marine sections of this summary. Eroded sediment carries nitrogen and phosphorus to coastal waters, where it drives the algal blooms that produce hypoxia. Soil conservation and coastal oxygen are the same problem viewed at two ends of a watershed.

6. Biodiversity loss

The IPBES Global Assessment (2019) remains the authoritative synthesis, with its widely cited finding that around one million species face extinction risk. Two more recent IPBES assessments — the Nexus Assessment and the Transformative Change Assessment, both approved at IPBES-11 in Windhoek in December 2024 after five years of work by over 250 experts from more than 50 countries — are less quoted and arguably more useful.

The Nexus Assessment's central finding is that existing actions fail because they address biodiversity, water, food, health and climate as separate problems when they are interlinked. The Transformative Change Assessment identifies three underlying causes of the biodiversity crisis: disconnection from and domination over nature and people; inequitable concentration of power and wealth; and prioritisation of short-term individual and material gain.

A caution on metrics. The Living Planet Index is the most frequently cited biodiversity statistic and the most frequently misread. Its aggregate declines are driven by a relatively small cluster of extreme population declines; it does not mean that a typical monitored vertebrate population has fallen by the headline percentage. Where population trend data are used, they should be characterised precisely.

7. Cross-cutting drivers

Four drivers recur across all four topic areas above:

Land conversion for agriculture is the primary proximate driver of both deforestation and soil degradation, and a major driver of biodiversity loss.

Nutrient loading links agriculture to coastal hypoxia and is one of the planetary boundaries assessed as most severely exceeded.

Climate change amplifies every other pressure — weakening carbon sinks, driving ocean deoxygenation, shifting species ranges faster than they can track, and increasing fire and drought frequency.

Fossil carbon combustion sits upstream of climate change and is separately the dominant anthropogenic oxygen flux. At roughly 10.3 GtC per year of fossil emissions against a remaining budget of about 50 GtC for a 50% chance of limiting warming to 1.5°C, it is the single largest lever in this document.

8. Interventions

This section states effect sizes where they exist and attaches the caveats that accompany them. Interventions are grouped by strength of evidence rather than by topic.

Interventions with credible causal evidence

Securing collective land tenure for Indigenous communities. The strongest quasi-experimental evidence in this document. A regression-discontinuity study using Brazilian satellite data from 1982–2016 across 245 Indigenous reserves found that territories receiving full property rights showed substantially reduced deforestation, with an average effect of around 66%; territories without full rights showed no such effect. A follow-up found that full property rights also increased secondary forest regrowth on previously cleared land, by about 5% under the main specification and 2.21% under a difference-in-differences approach.

Payments for ecosystem services. A randomised controlled trial in western Uganda offered forest-owning households annual payments of 70,000 Ugandan shillings per hectare across 121 villages, 60 of which received the programme for two years. Tree cover declined 4.2% in treatment villages against 9.1% in controls, with no evidence of displaced cutting nearby, and the value of delayed CO₂ release exceeded programme cost even assuming clearance resumed after payments ended.

This result should not be generalised without care. A published response to the study argues it is unlikely to be predictive of outcomes elsewhere: the trial ran in a context of unusually high deforestation driven by low-return activities, and it differs substantially in spatial extent and design from leading large-scale programmes. Allocating resources on the expectation of comparable results risks disappointment.

Legally binding nutrient reduction targets for coastal watersheds. The interventions that have measurably reduced coastal hypoxia have used specific numerical requirements rather than voluntary measures. This is the most tractable intervention in the ocean section.

Interventions that work, but by less than commonly claimed

Protected areas. Effective, but effectiveness is systematically overstated because protected areas are disproportionately sited on land facing low development pressure. Matching analysis of Costa Rica's system found roughly 10% of protected forest would have been cleared absent protection, and that conventional comparisons overestimate avoided deforestation by more than 65%. A Colombian study found deforestation inside protected areas about 40% below matched controls, varying sharply by region. An Amazon fire study using matching,

difference-in-differences and fixed effects found protection effective but by a margin 1.4 orders of magnitude smaller than naive estimation implied.

The implication is not that protected areas fail. It is that placement and enforcement quality matter far more than headline coverage percentages — currently about 20% of forests globally, ranging from 26% in Asia to 17% in South America.

Ranking

Two orderings are given, because they differ.

By oxygen relevance, as the original request specified: fossil fuel phase-out; protection of peatlands and coastal blue carbon ecosystems, which are the ecosystems that actually bury carbon and thereby produce net oxygen; coastal nutrient control, which directly addresses dissolved oxygen; forest protection, which is approximately oxygen-neutral; soil carbon management.

By overall importance to biosphere integrity: fossil fuel phase-out; tropical forest protection, weighted by carbon, hydrology and biodiversity together; coastal nutrient control; peatland protection; agricultural land-use efficiency, since agriculture is the proximate driver of most terrestrial loss.

Fossil fuel phase-out heads both lists, and by a wide margin on the first.

9. What the evidence cannot tell us

Most conservation interventions have never been evaluated counterfactually. Where they have, measured effects are consistently smaller than advocacy claims, primarily because naive comparisons mistake favourable siting for policy impact.

Three specific gaps are worth stating. Cost-effectiveness comparisons across intervention types barely exist; there is no conservation equivalent of a common effectiveness metric permitting comparison between, say, tenure reform and marine protection. Long-run persistence is largely unmeasured — most evaluations run for a few years, and permanence is the central question for carbon-motivated interventions. And leakage is inconsistently assessed; the Uganda trial found none locally, but displacement at larger spatial scales is harder to detect and less often looked for.

Conservation Evidence, based at Cambridge, maintains the most systematic effort to synthesise intervention evidence, with synopses covering forest conservation, peatland conservation and

soil fertility among others. Its own framing is appropriate: the evidence summarises what is known as a starting point for deciding whether to act, not as a verdict.

10. Sources and provenance

Forests: FAO, *Global Forest Resources Assessment 2025*. Carbon Brief analysis of FRA 2025. Global Forest Watch / World Resources Institute.

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Soils: FAO & ITPS, *Status of the World's Soil Resources* (2015). FAO Global Symposium on Soil Erosion (2019). FAO, *Global status, processes and trends in soil erosion*.

Biodiversity: IPBES *Global Assessment* (2019); IPBES *Nexus Assessment and Transformative Change Assessment* (2024). IUCN Red List. Leung et al. 2020, *Nature*, on Living Planet Index interpretation.

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Planetary boundaries: Richardson et al. 2023, *Science Advances*.